



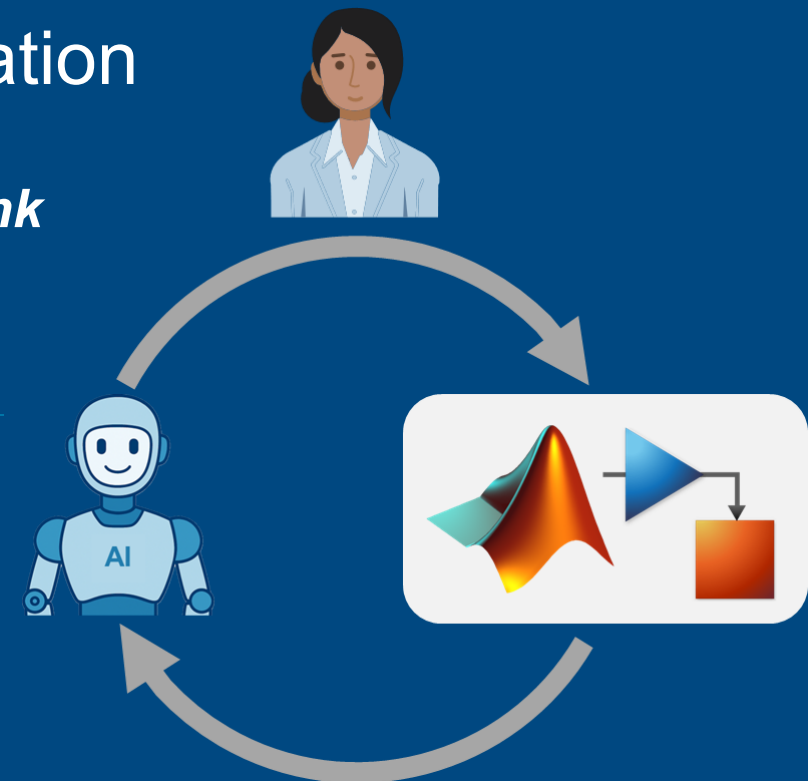
AP-S/URSI — Technical Workshop HD-7

AI in Antenna Design and Analysis: From Pretrained Models to System-Level Integration

Part 4 - Agentic AI Workflows with MATLAB and Simulink

Sekhar Sekharan— sekhars@mathworks.com

Applications Engineer
MathWorks | RF/EM Group

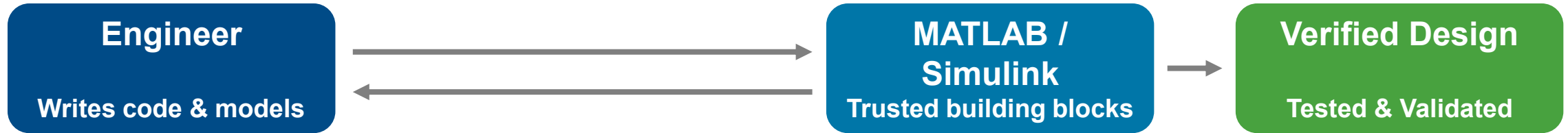


Generative AI is transforming engineering

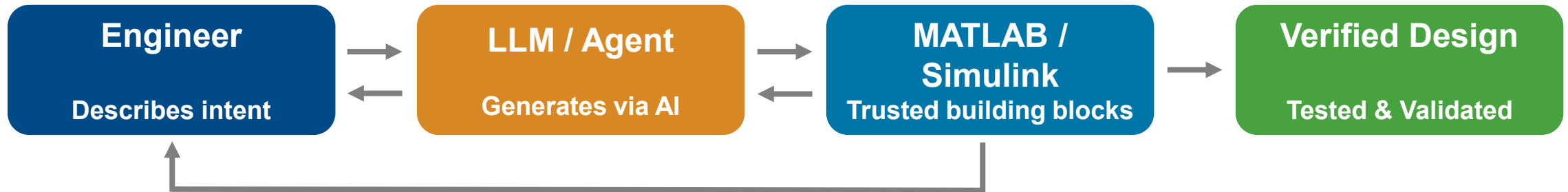


GenAI changes how you create - not engineering accountability

Today



With AI



What doesn't change when you add AI

The engineer owns the design. The same engineering evidence verifies it.

Accelerate your MATLAB and Simulink productivity with GenAI

Agentic Workflows



External GenAI Agents

*use MATLAB or Simulink with external Agents
via MCP or VS Code*



Build Agents in MATLAB

call LLMs from MATLAB code

AI Assistants



MATLAB and Simulink Copilots

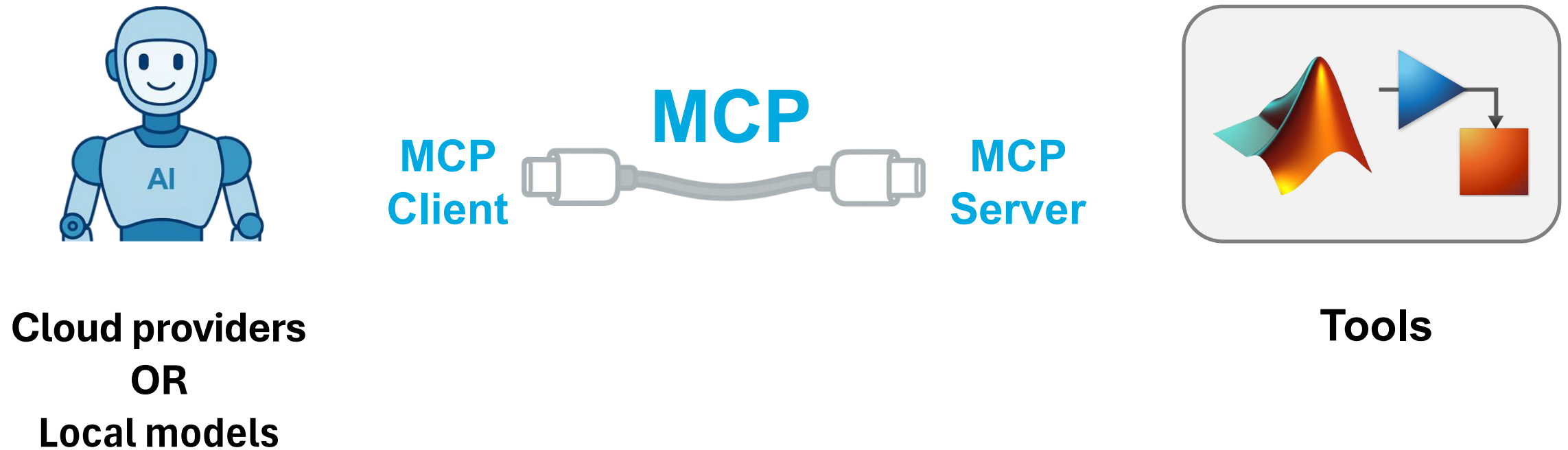
optimized for MATLAB and Simulink



External GenAI Agents

*use MATLAB or Simulink with external Agents
via MCP or VS Code*

Agentic workflows are enabled by Model Context Protocol (MCP)

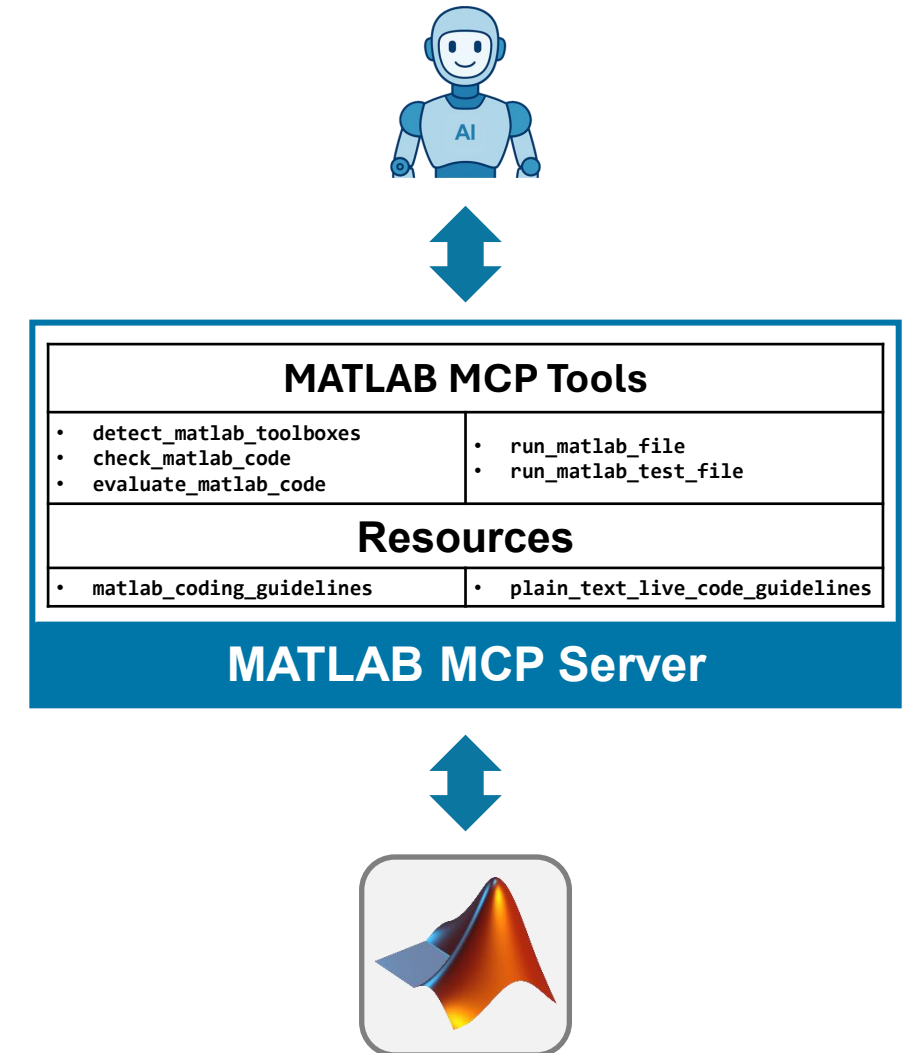


MATLAB MCP Server

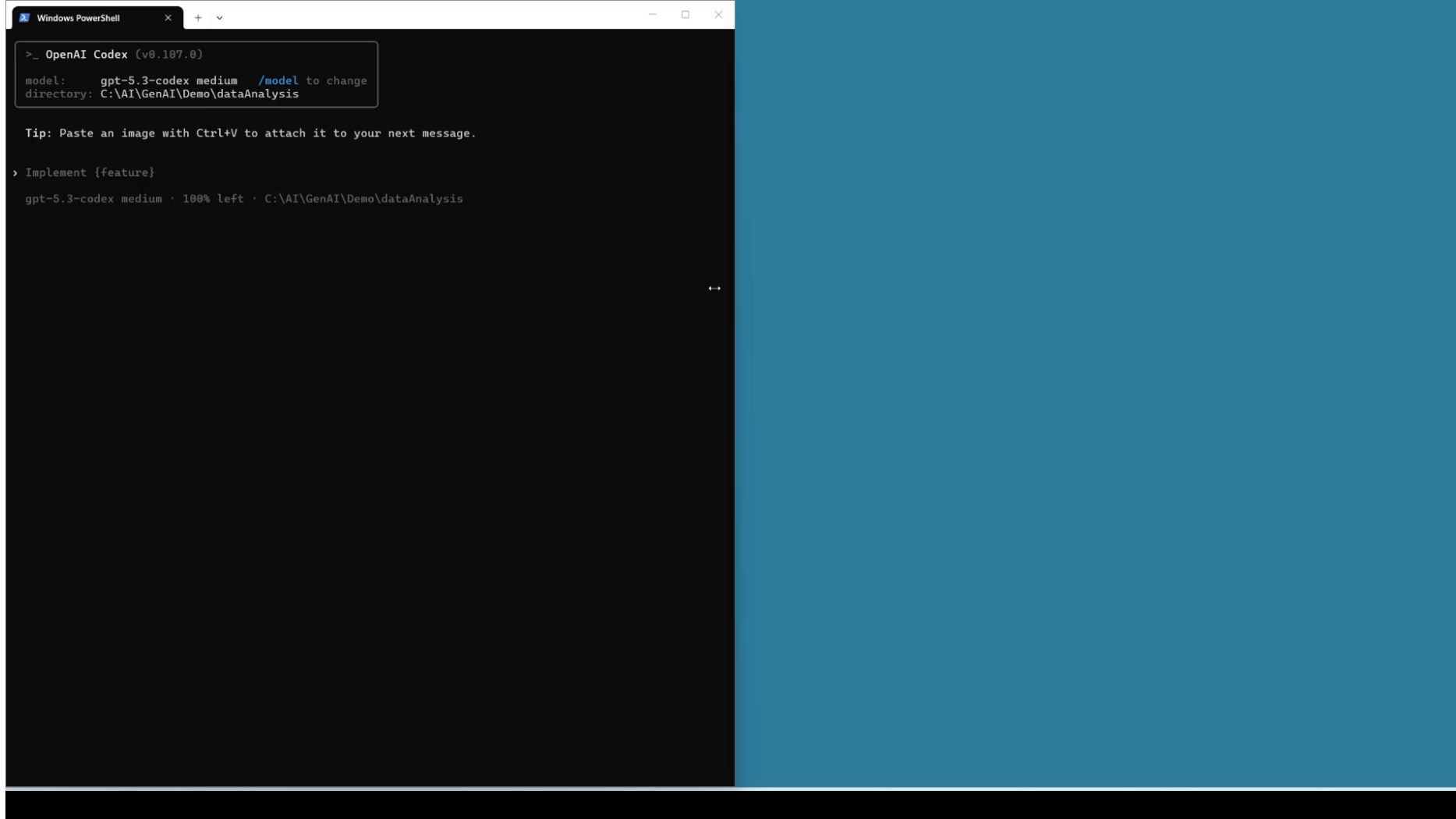
Enabling GenAI Agentic Workflows with MATLAB

A **free, open-source** local MCP server that connects GenAI assistants to a **locally** licensed MATLAB, enabling intelligent automation and code execution

- No additional license required – just connect and go
- Preserves privacy with local execution
- Start a new MATLAB session or connect to an existing session



Create and execute MATLAB code



```
Windows PowerShell
>_ OpenAI Codex (v0.107.0)
model:      gpt-5.3-codex medium  /model to change
directory:  C:\AI\GenAI\Demo\dataAnalysis

Tip: Paste an image with Ctrl+V to attach it to your next message.

> Implement {feature}
gpt-5.3-codex medium · 100% left · C:\AI\GenAI\Demo\dataAnalysis
```

How can we improve agent behavior?

Skills transform prompts into reliable, domain-aware, reusable workflows

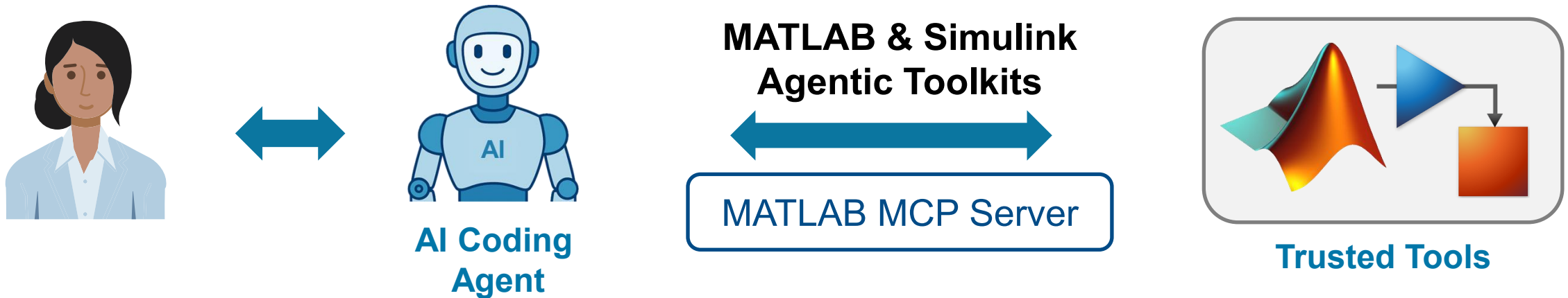
- Right knowledge at the right time
- Reliability and output quality
 - Encoding proven procedures
 - Grounding against hallucination
- Knowledge the model may not have
 - Domain and proprietary expertise
 - Institutional memory

/skills/skill-name/SKILL.md

```
---  
name: skill-name  
description: Defines the conditions under which  
the skill should or should not be triggered  
---  
  
# Instructions  
This section is loaded by the agent only if it  
determines, based on the skill description, that  
the content may be relevant to the current task.
```

Skills tells your agent how to do things correctly the first time around

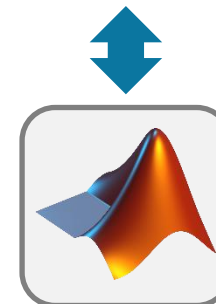
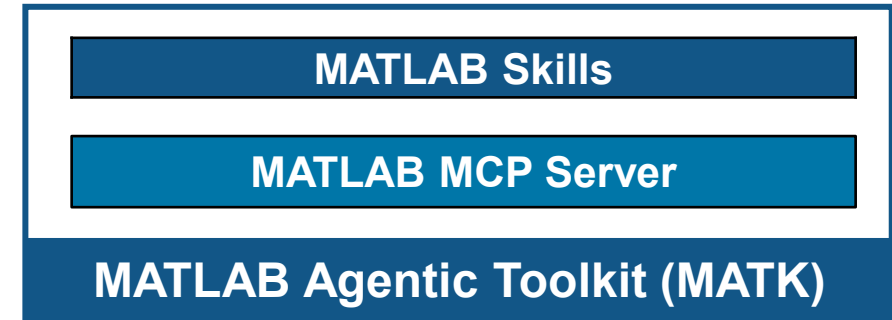
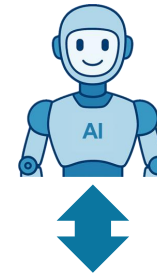
MATLAB & Simulink Agentic Toolkits: Bring Trusted Capabilities to AI Agents



MATLAB Agentic Toolkit

Connect trusted MATLAB capabilities to AI agents to make engineering and scientific workflows agent-ready

- MATLAB MCP Server
- Expert Skills Catalog encode MATLAB best practices, guiding AI in testing, debugging, and code modernization.
 - MATLAB Software Development
 - Signal Processing
 - Image Processing and Computer Vision
 - Wireless Communications
 - .. and many more!!



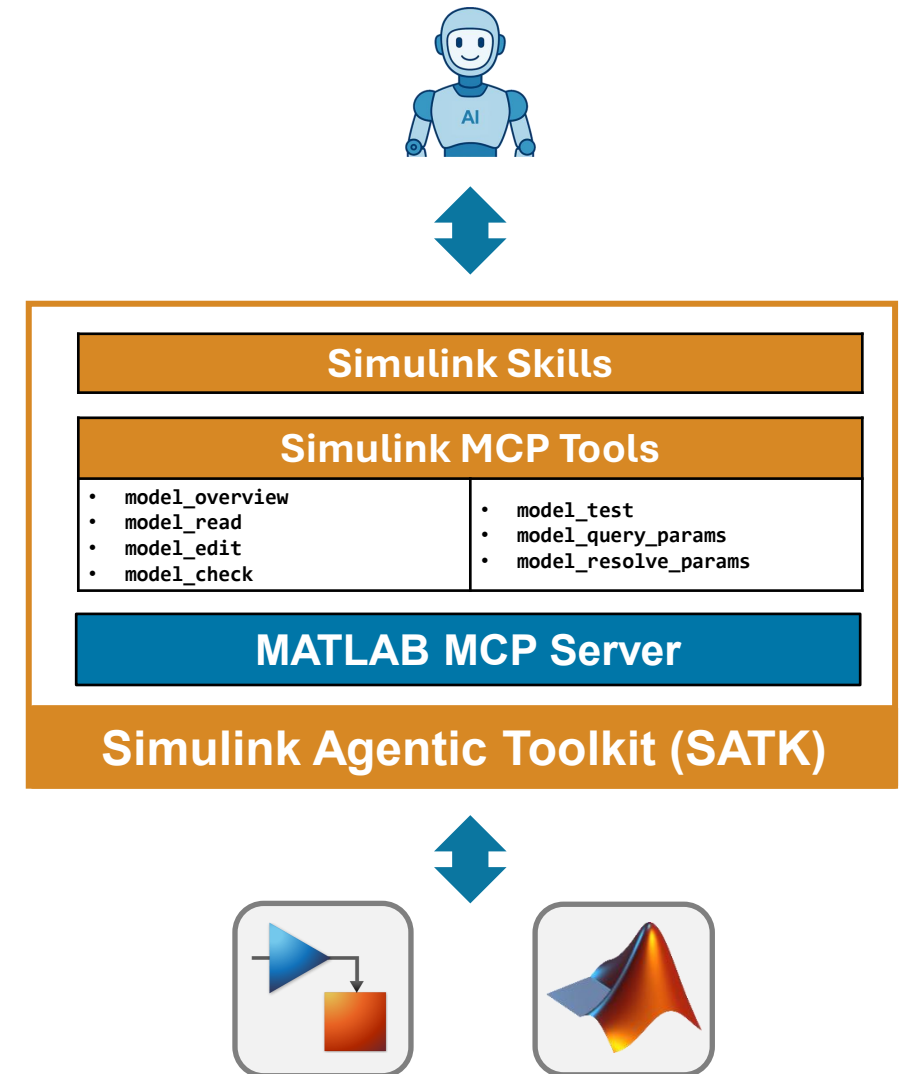
Demo: Antenna Design using Agentic AI

Simulink Agentic Toolkit

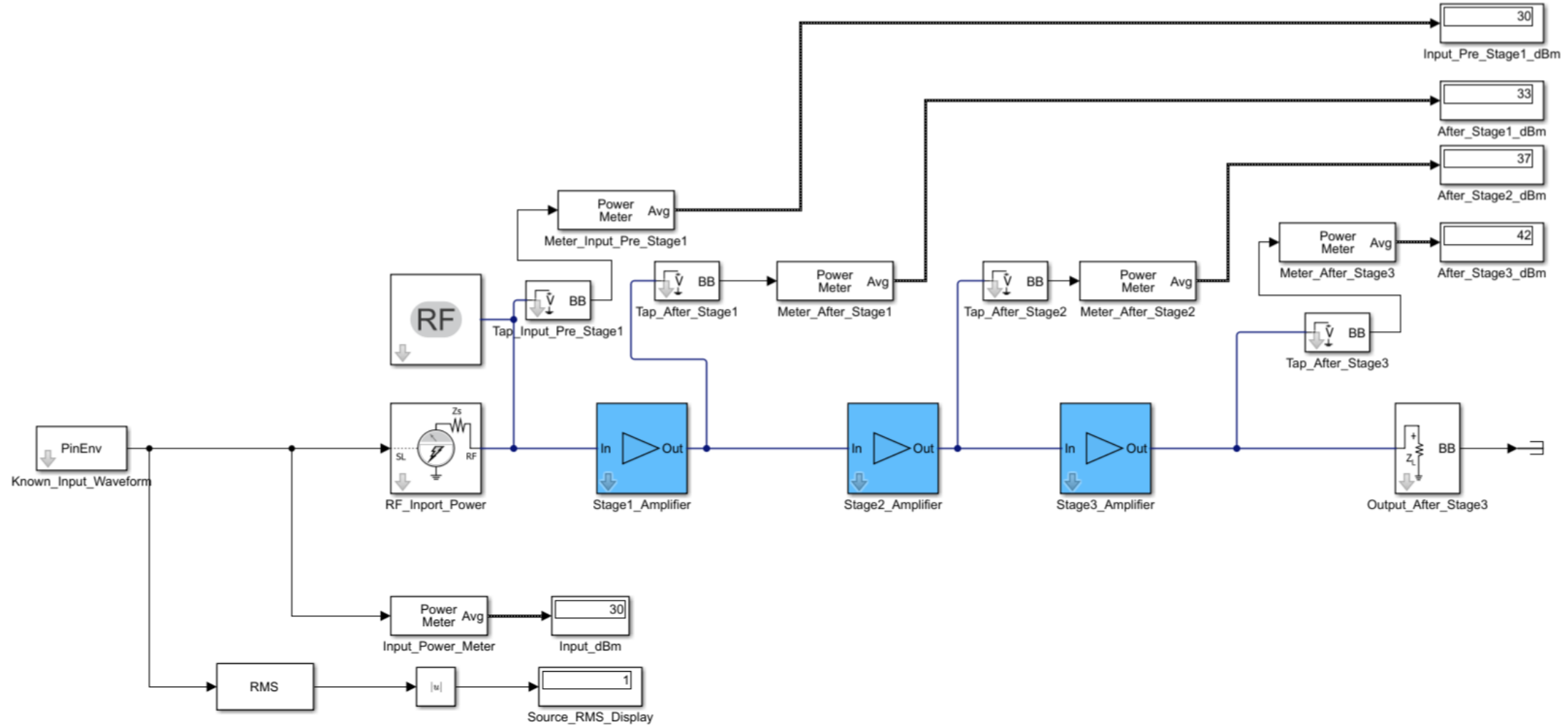
Connect agents to Simulink giving them the ability and the knowledge to work with Simulink models effectively.

- Builds on top of the MATLAB MCP Server
- MCP tools optimized for reading, editing, querying, testing, and checking Simulink models
- Agent skills encoding Model Based Design best practices for model building, plant specification, testing, requirements, and architecture modeling

<https://github.com/matlab/simulink-agentic-toolkit>

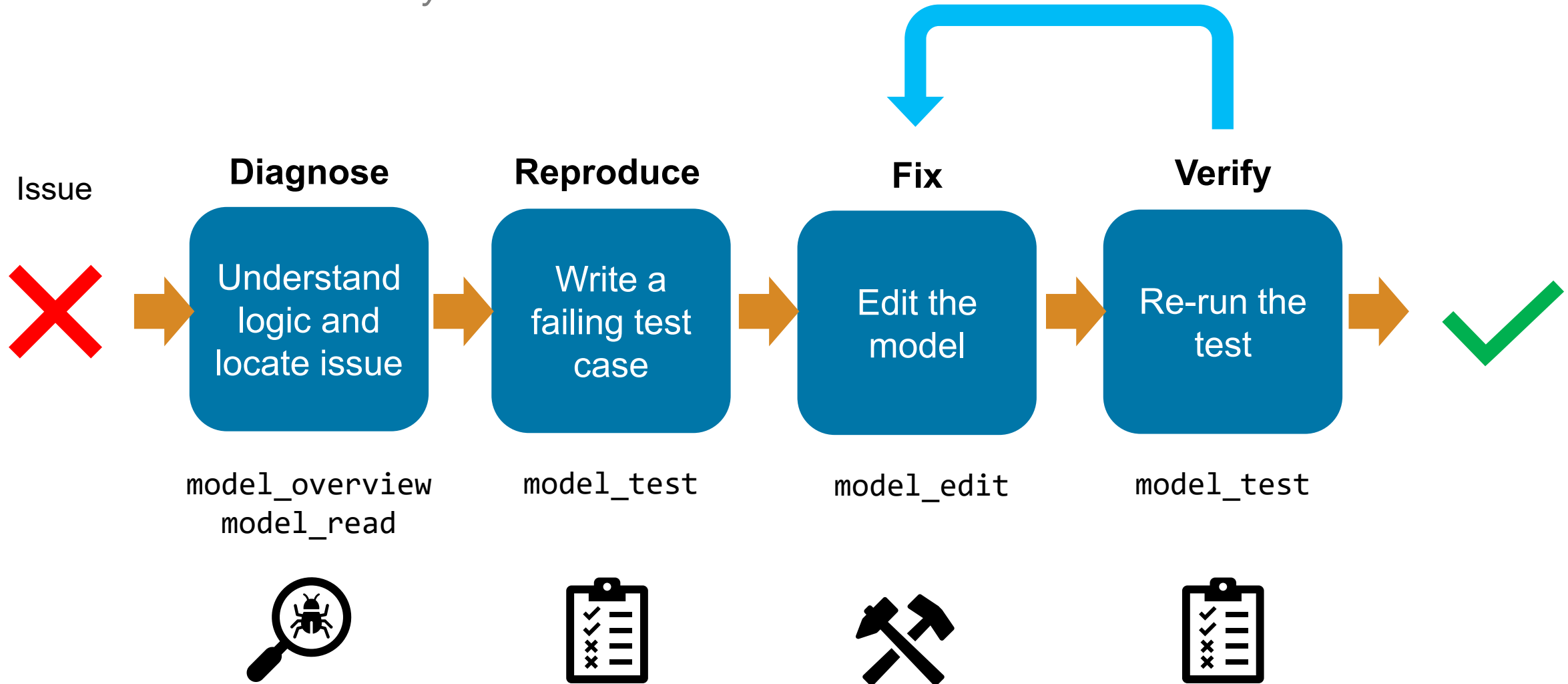


Example RF Model Generated with Agentic Workflow



Simulink Agentic Toolkit enables automation of complex workflows

Troubleshoot – Fix – Verify



Other offerings

- [Build custom agents in MATLAB](#)
- [Deploy your MATLAB algorithms and models as MCP tools](#)
- [Polyspace Agentic Toolkit](#)
- [MATLAB Copilot](#)
- [Simulink Copilot](#)
- [Polyspace Copilot](#)

Key Takeaways

- Agentic AI is powerful; with the right tools it is transformational.
- Regardless of your AI tools, there is a way to use them with MathWorks Tools
- Stay informed about new updates:
<https://www.mathworks.com/products/matlab/agentic-ai.html>
- Ready to get started? Schedule a follow up to discuss workflow challenges and explore new ideas

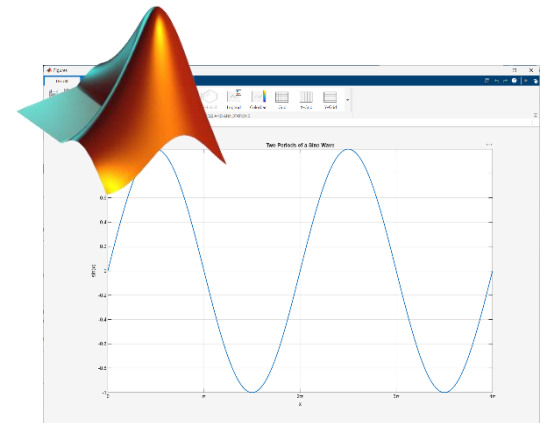
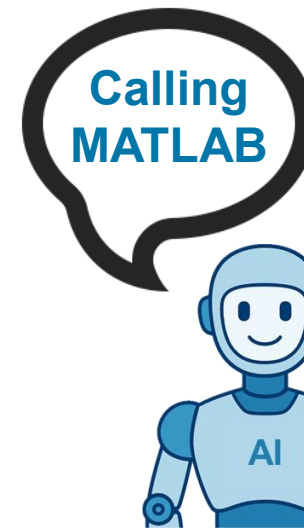
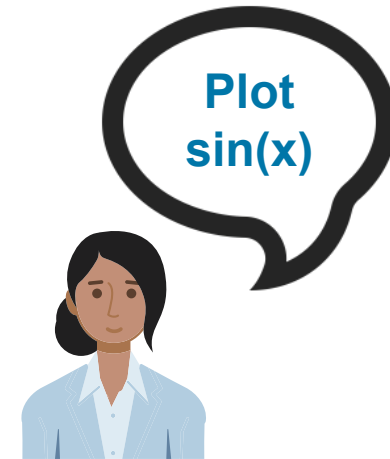
Resources

- [Generative AI with MATLAB and Simulink](#) (Overview)
- [MATLAB MCP Server](#)
- [MATLAB Agentic Toolkit](#) (MATLAB MCP Server + MATLAB Skills)
- [Simulink Agentic Toolkit](#) (MATLAB MCP Server + Simulink Tools/Skills)

Accelerating Design and Analysis in MATLAB Using Agentic AI

After this 1-day training you will be able to:

- Explain how agents operate and differ from past tools
- Connect MATLAB with AI agents
- Work with agentic AI to create MATLAB code and accelerate complex tasks
- Understand, debug, and extend large code bases in MATLAB with agentic AI assistance
- Leverage domain expertise by creating custom skills for repeatable workflows



Note: this training is available only for private training events

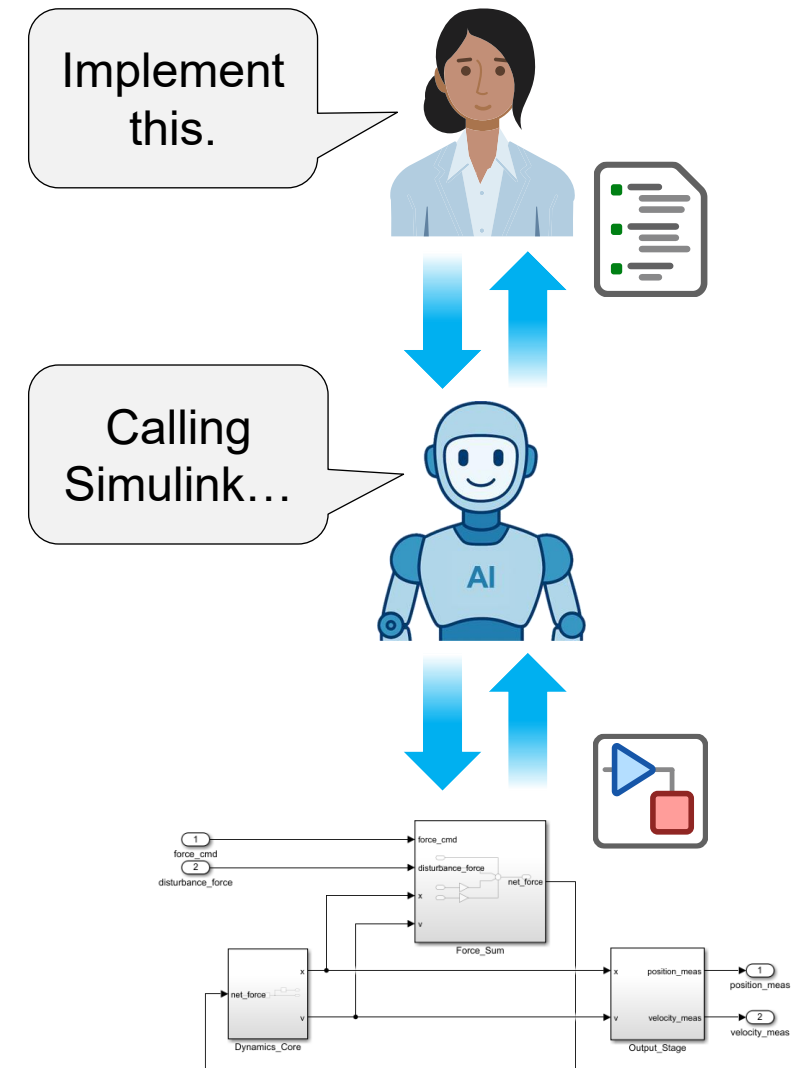
Accelerating Engineering Design in Simulink Using Agentic AI

After this 1-day training, you will be able to:

- Recognize how agentic AI systems use context, tools, and review loops to support Simulink engineering tasks.
- Build, inspect, and edit basic models using the Simulink Agentic Toolkit.
- Use agentic AI to complete a Simulink engineering task with review checkpoints, simulation, testing, and documentation.
- Improve consistency and incorporate domain knowledge for Simulink workflows using custom skills.

[Full Course Outline](#)

Note: this training is available only for private training events



MathWorks Engineering Support



Training



Guided Evaluations



Onsite Workshops



Consulting



Technical Support

Questions?